

Trading on the Information Content of Open Interest: Evidence from the US Equity Options Market

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Abstract

In this paper, we use daily closing data on CBOE options of 30 stocks during February through July of 1999 to investigate whether options open interest contains information that can be used for trading purposes. Individual stock price at option maturity is first predicted based on the distribution of options open interest. Several stock only and stock plus options directional trading strategies are then considered after comparing the predicted stock price at maturity and the actual stock price at the trade initiation date. In our sample, these trading strategies generate better returns compared to the S&P 500, the buy and hold strategy involving the sample stocks and the Merton et al (1978) style covered call strategy. Our empirical evidence thus indicates that non-price measures of activity in the derivatives market such as the open interest contain information about the future level of the underlying asset. This lends support to prior works (such as Copeland and Galai (1983), and Easley et al (1998)) that suggest that the derivatives cannot be considered redundant in a market with information-related frictions. One implication is that the distribution of non-price derivatives market activity may be helpful for other purposes where the physical instead of the risk-neutral distribution of the underlying asset is needed. These include beta estimation, volatility forecasting and volatility trading.

Keywords: information content, options, open interest, trading strategy.

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DO PRICES AND TRADING ACTIVITY in security markets provide information about future price movements? If so, can this information be used to generate trading gains? While these questions have long generated significant interest among researchers and practitioners alike, the existence and growth of derivatives such as options have added new and interesting issues to this arena. The derivatives constitute an additional means for the informed traders to trade on their information and others to discover that information. Not only the derivatives may lead the underlying assets in impounding information, they may in fact provide information that simply cannot be inferred from the markets in underlying assets. Further, there are measures of trading activity, e.g., open interest, that are unique to the derivatives markets and as such provide a novel way to examine the informational role of financial markets.

In this paper, we examine the role of options market open interest in conveying information about the future movement of the underlying asset. More specifically, we use the open interest of short term equity options to predict the stock price at maturity and show empirically that trading strategies based on this predictor yield better returns than the buy-and-hold and passive covered call strategies. Thus, according to our study, (unique measure of) activity in the equity options market seems to contain information about future stock price that can be exploited for trading purposes. In principle, this information can also be helpful in other applications where the physical or true distribution of the underlying asset is needed.

Financial economists have long been interested in the process of price formation when informed traders, uninformed liquidity (or noise) traders and market makers interact in the asset market.¹ With the introduction of options market, informed traders as well as liquidity traders have an additional means to meet their trading needs. In fact, informed traders may find the options market more lucrative than the stock market due to lower transaction costs, less capital outlays, higher leverage, limited loss potential and lesser trading restrictions (e.g., no up tick rule for shorting).² If informed traders do choose to trade in the options market, not only the option prices

¹ In an asymmetric information environment, informed traders may profit at the cost of noise or liquidity traders' loss (Copeland and Galai (1983)). Continued trading of the informed investors can, however, serve as signals to the other (uninformed) market participants who can learn the underlying information in a Bayesian fashion and trade accordingly (Glosten and Milgrom (1985), Easley and O'Hara (1987), Kyle (1985)). This possibility of multiple rounds of trade arises if the first trade of the informed traders does not instantly reveal the new information. Although the implications for the price paths, volume changes and trading strategies are different, information-motivated trading may also be driven by differential information (He and Wang (1995)) or differential interpretation of the same information (Copeland(1976)) by informed traders.

² Black (1975) first hinted to the attractiveness of the options market to informed traders.

and options market activity become relevant in impounding the information and its subsequent discovery,³ options could in fact lead the underlying stock in terms of price change and trading activity. An impressive literature has emerged researching this and other related issues although the empirical evidence appears inconclusive.⁴

While much attention has been paid to the information content of derivative prices, theoretical research on the information content of derivatives market activity about the future movement of the underlying asset is only beginning to emerge.⁵ This line of research is mainly due to Easley et al (1998). They present an asymmetric information based theory where the informed investors engage in specific trades in call and put options depending on the direction (positive or negative) of their information. The volume of these directional option trades conveys information to other market participants and thus ultimately impacts the underlying asset price. In their empirical

³ Under asymmetric information, information may flow from the option market to the underlying asset market. This has important implications for the underlying asset and its options. Grossman (1988) suggests that under asymmetric information traded derivatives are not the same as their synthetic counterparts due to their differential information content. While Detemple and Selden (1991) argue that information asymmetry may alter the hedging opportunities and as such impact the underlying asset price, Back (1993) examines the impact on option prices. Biais and Hillion (1994) show that the price volatility of the underlying asset may be affected by information asymmetry. Brennan and Cao (1996), on the other hand, show that in a noisy rational expectations equilibrium, option trades may not be information based.

⁴ Using the Black-Scholes model, Manaster and Rendleman (1982) jointly imply the stock price and volatility from the observed daily closing option prices and find support for the incremental information content of options based on returns from ex post and ex ante trading strategies. Using the Berkley transaction data on options, Bhattacharya (1987) also finds incremental information content of options although the trading benefits seem marginal. Vijh (1988) questions the ex post results based on daily closing prices due to the bias arising from the nonsynchronicity of closing stock and option trades and the bid-ask bounce. Anthony (1988) uses volumes of options bracketing the daily closing price and find that for 64% of the sample stocks the options volume led the stock volume in a Granger Causality sense. However, about 48% cases are statistically significant in both univariate and multivariate causality tests. Stephan and Whaley (1990), on the other hand, find stocks to lead their options both in terms of intra-day price change and trading activity. Chan, Chung, and Johnson (1993) argue that Stephan and Whaley's results are due to different price discreteness rules in the stock and option markets. Examining how option prices move with option trades, Vijh (1990) concludes that option trades are not information-driven while Srinivas (1993) attributes Vijh's results to a sample selection bias. Sheikh and Ronn (1994) attributes unique patterns of returns in the options market to information-based trading there. John, Koticha and Subrahmanyam (1993) and John, Koticha, Narayanan and Subrahmanyam (2000) show that the impact of options trading depends on the margin and liquidity constraints faced by the informed and uninformed or liquidity traders. Mayhew, Sarin and Shastri (1995) find that a reduced equity options writing margin increases the bid-ask spread of the optioned stocks and that the uninformed traders are more liquidity constrained than the informed traders. This evidence goes against the prevalence of information traders in the options market.

⁵ The general equilibrium analysis of Leisen and Judd (2001) provides insights as to how open interests are determined along with the option prices for various strikes in an incomplete markets setting. In their paper, agents have heterogeneous risk preferences but homogeneous probability beliefs about the underlying assets. As such, derivatives market activity (the distribution of open interests) is not informative about the future of the risky asset.

study of 50 stocks with most actively traded options on the CBOE during October and November of 1990, Easley et al (1998) find that indeed the volume of directional option trades leads the stock price changes although the total option volume has no such predictive power. This shows that the pattern of derivatives market activity may contain information about the future movement of the underlying asset.

It is worthwhile to look at some of the assumptions underlying the theoretical model of Easley et al (1998). *First*, informed traders are assumed to know the true value of the asset. Regardless of their risk preference, the informed investors would then choose an investment strategy that maximizes profit barring any wealth constraint. It is, however, more realistic that the informed investors instead acquire signals with noise about the true value. In that case, it is not clear if the best strategy as outlined by Easley et al (1998) would stay the same. In particular, if the informed investors were risk averse, the strategy with maximum expected profit might not be the best choice. Further, as mentioned by John, Koticha, Narayanan, and Subrahmanyam (2000), wealth-constrained informed traders would seek to maximize expected return instead of expected profit from trading. This leads to a change in optimal investment strategy when the wealth constraints such as those arising from margin requirements become binding.

Second, Easley et al (1998) assumes that all informed traders have the same information or signal and they interpret the signal in the same manner. As such, all informed investors undertake the same strategy (positive information or negative information). In practice, investors are more likely to have diverse (less than perfectly correlated) signals and/or differential interpretation of the same signal. Accordingly, we may have a distribution of positions by the informed traders within the same market (asset or options). Further, recent theoretical research (Back, Cao, and Willard (2000)) shows that the asset price process may be affected by the nature of signal distribution and competition among the informed traders.

Third, as is traditional in the asymmetric information literature, Easley et al (1998) assume away the impact of adverse movements in the broader market on the asset in question. Suppose the asset specific information is positive. However, if the asset is high beta, it may in fact go down with the market in case negative information or event hits the broad market. To guard against such possibilities, the informed investors might pursue limited risk rather than outright naked or speculative positions suggested by Easley et al (1998). For example, a covered call strategy involves selling some call options although the asset-specific information is positive. Accordingly, “sell” call option orders does not necessarily mean a “negative” information trade.

Fourth, in their empirical study, Easley et al (1998) implicitly assumes that all option trades are opening trades. In fact, some of the trades may relate to closing down previously opened positions by the informed traders. Instead of “positive” or “negative” information, if at all these trades might indicate no further directional information as the previous information has likely been reflected in the asset price.

Lastly, while Easley et al (1998) and the extant research have primarily considered information about the future level of the underlying asset, Cherian and Vila (1997), Cherian (1998), and Cherian and Weng (1998) argue that investors may also have private information about the volatility of the underlying asset. In that case, they can directly trade on their information only in the options market. The identification of option orders with the “direction” of information in the Easley et al (1998) study is thus confounded by the volatility trades in options. Also, the pricing of options by the market maker and thus the determination of the best strategy by the directional trader may be affected. Moreover, since volatility-information can only flow from the options to the underlying asset market, the study of Easley et al (1998) and others are biased towards finding the options market to lead the underlying asset market.

In principle, private information may concern not just the mean and volatility but other higher order moments or the entire true probability distribution of the underlying asset when this distribution cannot be adequately captured by the lower order moments. Theoretical work to formalize such complex informational environments remains a daunting task, especially in the area of option pricing. At a practical level, however, it seems reasonable that options market activity should reflect the trading and thus belief of the variously informed traders about the future movement of the underlying asset. This is the fundamental premise of our empirical study of the information content of options open interest. While we do not provide any theory in this regard, we hope our empirical evidence will stimulate further research in this area.

The contribution of our paper can be summarized as follows. *First*, to our knowledge, this is the first study to examine the information content of the distribution of options open interest. We use the discrete distribution of equity option open interest across various strike prices as a proxy for the true or physical distribution of the stock price at option maturity. Since open interest reflects the accumulated open positions of variously informed traders, our method empirically generalizes the trading setup from information about one or two moments to consensus belief about the entire distribution of the asset.

In this paper, we focus on directional trading strategies alone for a learner investor who does not have any private information. Based on the distribution of options open interest for a given maturity, the learner investor predicts the stock price at that maturity. A comparison of the current stock price and the predicted price then leads to trading strategies involving the stock and its options.

We find that the open interest based price predictor has good accuracy. The prediction accuracy is not due to accuracy only in specific option month or trade initiation day. Importantly, in our sample, the open interest based active trading strategies generate better returns compared to the passive benchmarks (buying and holding the stock, investing in the S&P 500, and pursuing Merton et al (1978) type covered call strategy). The magnitude of the return advantage seems too high to be nullified by any risk disadvantage there may be due to the use of a rough proxy for risk in this paper. Our empirical evidence thus suggests that equity options open interest contain information about the future movement of the stock price.

Second, the information implied from derivative prices is about the risk-neutral distribution of the underlying asset. This information is certainly useful in many ways (such as hedging, pricing other derivatives of the same asset, identification of misvaluation, etc.). However, in many other applications (e.g., portfolio optimization and performance evaluation, estimation of beta or other higher order measures of risk, estimation of cost of capital, etc.), it is the true or physical distribution of the asset that is of interest. Since we only observe a single price of the asset at a time (or the bid-ask), traditionally researchers and practitioners use some form of time series data on the asset to estimate its physical distribution.⁶ This approach is reliable and economical when the underlying distribution shows negligible time variation. Further, if the underlying asset price does not move very much, then it becomes statistically challenging to estimate the underlying distribution based on the time series of the asset price.

The open interest based approach offers an alternative means to estimate the underlying physical distribution of the asset. Potentially rich set of cross-sectional data on derivatives market activity (options open interest and volume of various strike prices and maturity) that is available simultaneously at a given point in time can be used for this purpose. The reliability of this approach of course depends on the information content of derivatives market activity. Our empirical evidence in this regard is quite positive and as such we are hopeful that our approach can be enhanced to

⁶ Converting the implied risk-neutral distribution into a physical distribution generally requires preference specification.

further improve the estimation of the true distribution from derivatives market activity. For one thing, our approach is quite flexible in terms of updating the distribution. It is also non-parametric and model-free in that no model specification is needed for the underlying asset dynamics or for the derivatives price dynamics.

Third, for traders who decide not to engage in private information acquisition, we offer a new set of profitable trading strategies that rely on learning (almost free of cost) from the derivatives market (and not prices). The profitability of course depends on the informed traders deciding to trade in the derivatives market. Recent research lends support to such behavior of informed traders. Further, unlike Easley et al (1998) or John, Koticha, Narayanan, and Subrahmanyam (2000), our trading strategies allow a choice of the strike price for option trading. This seems more appropriate if the informed investors and our learner investor are wealth-constrained and seek to enhance expected return from trade. In this case, options may be favored due to their greater leverage potential and the choice of the strike price may accommodate various degrees of leverage desired by differentially informed or wealth-constrained investors. In addition, limited risk strategies such as covered call writing may help disguise information trading to the market makers who watch orders in both stock and option markets.

The rest of the chapter is organized as follows. In Section I, we present the model for open interest based stock price predictor. This is followed by description of the data and the methodology in Section II. In Section III, we outline the various trading strategies and the associated payoff patterns for our hypothetical learner investor. The empirical evidence on the comparative performance of the various passive and active strategies are then presented in Section IV. Summary and conclusions are drawn in Section V.

I. The Prediction Model for Stock Price at Maturity

Consider an equity instrument, or stock, for which there is a set of call and put options maturing at T , the current time being T_0 . Let the price of the stock at time t be S_t . Let $\{X_i, i = 1, 2, \dots, K\}$ be the set of strike prices for call options and $\{X_l, l = 1, 2, \dots, L\}$ be the set of strike prices for put options. The payoff at maturity to the buyer of a call option with a strike price X_i is $P_i^c = \text{Max}[0, S_T - X_i]$. For a strike price X_l , the put option buyer's payoff at maturity is $P_l^p = \text{Max}[0, X_l - S_T]$.

$S_T]$. For any $t \in [T_0, T]$, let O_{it}^c be the net open interest at time t of call options with the strike price X_i and similarly O_{it}^p be the time t net open interest at strike price X_i for put options.

The key premise of our work is that investors take option positions driven by their trading needs. These needs are dictated by their beliefs about future equilibrium stock price or its physical distribution based on private and public information available to them at the time.⁷ The nature of option market activity should thus contain information regarding consensus about future stock price movements, in particular about the distribution of stock price at maturity. The consensus, however, reflects the opinions of traders who chose to trade in the options market to meet their specific trading objectives.

It is possible that the option prices are also materially and systematically (by strike or maturity) affected by the option trading needs of the investor population as typically is the case in incomplete financial markets.⁸ Options then cannot be considered as *truly derivatives* of the underlying stock. This is because when non-redundant options are traded, they may provide information about the underlying stock and as such end up affecting the stock price and its dynamics.⁹ Consequently the standard option pricing models that assume invariance of the stock price and its dynamics to options trading do not apply.¹⁰ An important implication is that the practice of equating the model

⁷ We thus visualize the financial markets environment to be characterized by possible combinations of the following: asymmetric information, differential private information or differential interpretation of same public information, learning by the uninformed traders, hedging or insurance needs, and learning by the uninformed traders.

⁸ Detemple and Murthy (1994) shows that in an economy with heterogeneous but rationally updated beliefs, while the structure of intertemporal general equilibrium asset prices remain the same as under homogeneous beliefs, the price levels and their dynamics are indeed affected by heterogeneity of beliefs. Detemple and Selden (1991) considers a one-period general equilibrium model with a risky stock and two classes of investors with diverse beliefs about the risk of the stock payoff. They show that when option is introduced, it is traded (bought by high risk perception investor and sold by low risk perception investor) in a noisy rational expectations equilibrium and ends up changing (increasing) the stock price as the investor-specific demand for the stock changes. A further informational effect on the stock price is generated as the investors condition their beliefs on the option price as well.

⁹ For example, Back (1993) finds that the perceived conditional density of the terminal stock price may be bimodal under asymmetric information. Back considers the impact of information asymmetry in a market microstructure model of Kyle (1985) to include a call option. With one risk-neutral informed trader, some uninformed liquidity traders and market makers, options trading creates a stochastic volatility framework leading to an imperfect correlation between the stock and the option. Not only Black-Scholes type option pricing no longer prevails, option orders in fact convey different information than stock orders. For example, a call option buy order increases the probability of option expiring in-the-money disproportionately more than a stock buy order

¹⁰ An extensive empirical literature exists on the effect of equity options introduction on the underlying stocks. According to Conrad (1989), Detemple and Jorion (1990), Kim and Young (1991), earlier equity options introductions led to an increase in the price of the optioned stocks. Such benefits are, however, not visible for later option listings and may in fact have reversed into negative impacts (Sorescu (2000)). A number of studies (e.g., Conrad (1989), Skinner (1989), Damodaran and Lim (1991)) find option listings to stabilize the variance but having no impact on the beta. The variance effect in the US market is by no means uncontested. In the Canadian market, Elfakhani and Chaudhury (1995) find a variance and beta stabilization effect of the earlier option listings; the

prices to the observed option prices and inferring information about the consensus distribution of terminal stock price becomes questionable. Further, implying the risk-neutral distribution of terminal stock price along the line of Rubinstein (1994) is not quite informative about the physical distribution of the terminal stock price. In particular, the lack of implied information about the first moment, namely the expected terminal stock price, is limiting in the context of exploring the informational role of options in discovering the expected future equilibrium stock value and in designing information-based trading strategies.

In contrast, metrics of option market activity such as the net open interest are not restricted by the limitations of option pricing in a complex economic environment and yet they reflect diverse traders' option trading needs and as such their consensus beliefs about the underlying stock's physical distribution. One limitation of using non-price measures is that theoretical work has only begun to emerge to explain equilibrium option market activity such as the net open interests of various strikes.

In this paper, we propose, by conjecture and not by proof, that the distribution of net open interests at time t over the range of tradable strike prices for options maturing at T be used as a proxy for the consensus physical distribution of S_T , the stock price at maturity.

Define a call option-open interests-based predictor (COP) by:

$$S_t^C \equiv \sum_{i=1}^K X_i q_{it} \quad (1)$$

where $q_{it} = \frac{O_{it}^C}{\sum_{i=1}^K O_{it}^C}$.

The weight, q_{it} , attached to a given strike price, X_i , of a call option, is the net open interest of that strike price relative to the aggregate net open interest of all call options at time t . Since these weights are by definition between 0.0 and 1.0 and sum to 1.0, they can be construed as probabilities. The price predictor COP can thus be viewed as an expected terminal stock price where the expectation is with respect to the time t discrete distribution of the open interests for all call options

variance effect however reverses following the later option listings. While Conrad (1989) and Kim and Young (1991) find no impact of the US listing of put options on the optioned stocks, Elfakhani and Chaudhury (1995) and Chaudhury and Elfakhani (1997) report Canadian evidence that supports a risk reduction effect associated with put option listings.

on the stock maturing at time T . If the consensus physical distribution represents (likely noisy) rational expectation of the stock price at T , the price predictor should be a good predictor if the time t open interest distribution tracks the time t consensus physical distribution well, at least in terms of the first moment. Our premise in this paper is that this should be the case if the options market activity contains information about future stock price movement.

In a similar vein, a put option-open interest-based predictor, S_t^P , is defined as:

$$S_t^P \equiv \sum_{l=1}^L X_l q_{lt} \quad (2)$$

where $q_{lt} = \frac{O_{lt}^P}{\sum_{l=1}^L O_{lt}^P}$

The weight, q_{lt} , attached to a given strike, X_l , of a put option, is the net open interest of that strike price relative to the aggregate net open interest of all put options at time t .

Since financial markets may not be complete and information-related imperfections may be prevalent, a put option and a call option with similar terms may not be the mirror image of each other. Hence, open interests of both put options and call options may convey information about the terminal stock price. Accordingly, we derive a third predictor, CWOP, combining the open interests of both call and put options:

$$S_t^Z \equiv \frac{\sum_{i=1}^K O_{it}^C X_i + \sum_{l=1}^L O_{lt}^P X_l}{\sum_{i=1}^K O_{it}^C + \sum_{l=1}^L O_{lt}^P} \quad (3)$$

Assume that the positive private information of a group of informed investors reflect a more favorable distribution of the stock price at T leading to a higher (than the current price S_t) expected equilibrium stock price, $E(S_T)$.¹¹ Since their information is only probabilistic, one likely speculative strategy is to establish buy positions in out-of-the-money options of various strikes.¹² The more

¹¹ For simplicity, we assume risk neutral investors and a zero risk-free rate.

¹² Throughout this paper, we assume that there are competitive risk-neutral market makers in both stock and options markets.

optimistic the informed investors are, the distribution of their buy positions will shift further to higher strikes (deeper out calls). This will tend to increase the weight for higher strike call options. An alternative option strategy for the optimistic informed investors is to sell in-the-money put options. This will tend to increase the weights for higher strike put options. In either case, higher weights for the higher strikes will result in a higher price predictor, consistent with the optimism of the informed investors.

Further, since the information is only probabilistic and new information later may negate the earlier positive information, the optimistic investors may decide to hedge against potential downside movement by buying some out-of-the-money (strike lower than the current stock price) put options. Higher the risk of information reversal and greater the perceived potential shift towards poorer prospects, the insurative put buying will tend to attach more weight to the lower strike put options and will end up attenuating the price predictor.

Since we do not expect the optimistic informed investors to take positions in in-the-money call options,¹³ we should still end up with a positive bias in the price predictor due to the combined higher weights for the higher strikes (bought out-of-the-money calls and/or sold in-the-money puts) as opposed to the lower strikes (bought insurative out-of-the-money puts).

Now consider another group of informed investors whose information happen to be negative or who interprets the same information negatively. A likely speculative strategy for these pessimistic informed investors is to buy out-of-the-money (lower strike) put options. The more pessimistic they are, the greater will be their influence on the weight of the deeper out-of-the-money put options. Alternatively, they may also sell in-the-money (lower strike) call options. In either case, their open positions will tend to lower the level of the price predictor, consistent with the negative information of this group of investors. Of course, these investors also face the risk of information reversal and may hedge against a sharp run up by buying deep out-of-the-money (higher strike) calls. Since the pessimistic informed investors are not likely to sell in-the-money (lower strike) put options, their net influence is expected to be negative on the price predictor via the combined weights of lower strike put options bought and/or lower strike call options sold.

In addition to the informed investors, optimistic and pessimistic, there are liquidity traders and uninformed speculators. Their option actions are expected to be random and hence constitute a source of forecast error for the price predictor. Since information arrival can be sequential, it is also

¹³ For most optioned stocks, margin buying already represents some leverage. As such, in-the-money call options are not likely attractive to the optimistic informed investors who choose to trade in the options market.

possible that the open interests reflect positions undertaken by learners. The learners, however, have to infer the potentially conflicting signals of the optimistic and the pessimistic informed traders. Their inference errors would also add noise to the option market activity and thus the price predictor.

It is the perspective of a new learner that we assign to our hypothetical investor in this paper. This learner attempts to learn from the publicly available distribution of the open interests over various strike prices, constructs the price predictor, S_t^Z , and trades accordingly either in the stock market alone or both in the stock market and the options market. These trading strategies are delineated in the next section and their simulated performance is evaluated thereafter.

II. Data and Methodology

A. Data

The daily closing call and put options data (price and open interest by strike) used in this study was collected from the online delayed quote reporting system of Dreyfus Brokerage Services (DBS). These quotes are derived from the Market Data Report of the Chicago Board of Options Exchange (CBOE).¹⁴ Our sample period spans the six consecutive option months of February, March, April, May, June, and July of 1999. We define an option month as the period between the two consecutive option expiration dates. For example, the February option month extends from the first day of trading after the options expiration date in January to the last day of trading before the options expiration date in February.¹⁵

Our sample consists of thirty popularly held companies chosen from the NASDAQ and the New York Stock Exchange (NYSE). A list of these companies and the sectors that they represent are provided in Table I. These companies were selected to represent major market indexes, a cross section of important sectors and active options trading on the CBOE. The distribution by index is as follows: DJIA (11), S&P 500 (7), NASDAQ 100 (7), NASDAQ Composite (3) and DJTA (2). The sample firms represent seven broad sectors: technology (11), services (10), consumer products (3), basic materials (2), conglomerates (2), energy (1) and healthcare (1).

¹⁴ We cross-checked the DBS data against the CBOE data available online and found no major discrepancies.

¹⁵ The options expiration date in any calendar month is usually the third Friday of the calendar month. In case, the third Friday is a holiday, we use the last trading day before expiration as the last day of the option month.

Table II gives some statistical information about our sample firms such as the market capitalization, shares outstanding, average daily trading volume and market beta. Betas ranging from 0.70 (Du Pont) to 2.59 (Applied Materials) show ample variation in market risk across the sample firms. In terms of market capitalization, firm size ranges from \$5.93 billion (Altera) to \$359.40 billion (General Electric). Shares floating versus shares outstanding show that the sample firms have wide circulation for trading purposes. Technology bell weathers Dell (44.20 million), Compaq (16.00 million) and Oracle (15.60 million) were the three most actively traded stocks in our sample.

B. Methodology

Our active investor (learner) may choose one of the following four types of basic strategies with respect to a given stock: passive buy-and-hold (stock only), Merton et al type covered call (stock plus option) strategies, open interest based active strategy using only stock, and open interest based limited risk active strategies using stock plus options. For each type of strategy, equally weighted portfolio returns are calculated averaging the returns from positions with respect to the individual stocks in the sample.

For all strategies, positions are established using the closing prices of a trade initiation day (t) within an option month and the positions are liquidated using the closing prices on the last day of trading (T) during the option month. Thus, options that are permissible for trading have less than thirty calendar days to expiration. Within an option month, we allow trades to be initiated on four trading days: the second Monday (2M), the second Friday (2F), the Friday before the expiration Friday (LF) and the last Monday, i.e., the Monday of the expiration week (EXM).¹⁶ Thus, the corresponding holding periods in terms of calendar days are about 20 days (2M), 14 days (2F), 7 days (LF) and 4 days (EXM).

We chose to study short (less than a month) holding periods since information nowadays circulate quite rapidly due to the explosive growth of internet usage specially among the investing population. As such, learning has become easier and faster and information based trading is expected to impound the information into prices without significant delays. Also, active traders tend to have a short horizon. The 2M trade initiation day is chosen to allow about a week of option trading during the option month following the last expiration, so it allows a week of learning or

¹⁶ As is customary, if a Monday is a holiday, trading is initiated using the closing prices of the next available trading day. Similarly, if Friday is a holiday, positions are liquidated using the closing prices of the immediately preceding trading day available.

digesting the option market activity information by the active traders. The 2F is midway through the option month while the LF allows the option market activity to start reflecting the trades of informed traders based on the large amount of macro, industry and company statistics that are typically compiled around the turn of the month. Lastly, we include the EXM trade initiation day to pick up any information-based peculiarity of the weekend before expiration or potential last minute strategic trading moves by the market participants.

We allow a sum of around \$10,000 to be invested in a given stock and/or its options on a given trade initiation day. Since an option contract is for 100 shares, we had to undertake some rounding (to be discussed later) to determine the number of contracts and shares that could be traded. As a result, the investment sum for a stock (I) could be different from \$10,000 in some strategies involving stock and options. Once the positions are established based on the distribution of open interests and using the closing stock and option prices for trade implementation, they are held till the end of the option month.

Under each strategy, for each stock, *first*, the dollar profit or return (P) is estimated using the closing stock and option prices on the trade initiation day and the option maturity date. In calculating the dollar return we assume a zero transaction cost for the stock trading, \$30 for one-way option trading and zero transaction cost for option exercises. *Second*, for each stock, the holding period percentage return on investment ($PROI$) for the strategy is calculated based on the dollar return and the estimated investment (around \$10,000). The performance of a strategy is then measured as the (cross-sectional) average of the percentage return on investment with respect to the thirty sample stocks. Whenever possible and appropriate, we also present the cross-sectional standard deviation of $PROI$ as a proxy for the risk or volatility of returns on a strategy.

Further, for a small number of stocks, there were major news events after the trade initiation day (prior to the next option expiration date) during our sample period. Following these events, the stock price often moved significantly and seemingly against recent stock trend. When this happens, the expiration day actual stock price may end up being widely different from the open interest based predictor. If prior to the news event, the stock was trending up and the open interest based predictor was also signaling an upward move, then it is more reasonable to assume that any negative information pertaining to the news event came as a surprise to all investors.

To see the impact of these news shocks on the relative performance of the active strategies, we estimated two sets of results for the open interest based active strategies. One set of results assume that our hypothetical investor did not have a clue that news shocks are coming for some of

the stocks and hence established positions in all thirty stocks using the pre-news open interest based predictor and the pre-news closing prices on trade initiation day. We call this portfolio the IN (Ignore News) portfolio. The second set of results assumes that our hypothetical learner investor had guessed correctly the stocks for which the news shocks were coming. However, our investor did not know the nature (positive or negative) of the news shocks. Since we restrict the investor to directional trading only, in the second set of results we assume that our hypothetical investor did not establish positions with respect to the stocks on the trade initiation day. The performance of an active strategy in this case is the average PROI on the positions with respect to the remaining non-event stocks. We refer to this portfolio as the CN (Consider News) portfolio. Of course, we would expect the PROI to be higher for the CN portfolio.

It turned out that even for the longest holding period in our sample, only one to two of the thirty sample stocks had major news events that led to a significant divergence between the performance of the strategy (for the stocks in news) ignoring and considering the news. Considering the portfolio of all the stocks, the performance ranking of the various strategies remained by and large the same. Further, we only consider a sample of thirty stocks. For active institutional investors, it is likely that the portfolio would involve many more stocks and as such portfolio performance perturbations due to stock-specific news events would tend to average out. Therefore, while we present some evidence regarding the effect of the news events, most of the empirical evidence for the active strategies is presented using the CN portfolio. The generic name that we use for the active portfolios in presenting the results is OPP. If not mentioned otherwise, OPP would thus represent the CN results in the empirical evidence section.

III. Trading Strategies

In this section, we shall delineate the specifics of the trading strategies with respect to a given stock that our hypothetical learner investor may follow. These strategies are respectively the passive buy-and-hold (stock only) strategy, Merton et al type covered call (stock plus option) strategies, open interest based active strategy using only stock, and open interest based limited risk active strategies using stock plus options.

A. Passive Buy and Hold Strategy (Stock Only)

For a given stock, the buy-and-hold strategy simply involves buying shares at the closing price of the trade initiation day. The number of shares bought is determined by dividing the total dollar amount to be invested (\$10,000) by the closing stock price on the trade initiation day and rounding to the closest integer if necessary. It is assumed that investors are not allowed to borrow securities on margin, and the risk free interest rate is assumed to be zero as the interest up to three weeks is inconsequential. We do not consider brokerage fees for stock trading since many discount brokers charge as low as ten dollars or less.

The expected dollar returns for the buy-and-hold strategy for a stock is:

$$\begin{aligned}\Pi_{BH} &= [(S_T - S_t) \times N_s] - I \\ N_s &= 10,000 / S_t\end{aligned}\tag{1}$$

where, S_T is the closing stock price at the option maturity date, S_t is the closing stock price on the trade initiation day, N_s is the number of shares bought, and I is the net outlay. Here, $I = S_t \times N_s \cong 10,000$. The percentage return on investment (PROI) is estimated as follows:

$$PROI_{BH} = (\Pi_{BH} \div I) \times 100\tag{2}$$

An equally weighted portfolio of all thirty stocks is then formed. We call this portfolio the naïve investor or NI portfolio. The holding period return on this portfolio is simply the average return on the thirty stocks from the specific trade initiation day to the end of the option month.

B. Merton et al (Passive) Covered call strategy

A covered call strategy differs from a buy-and-hold strategy in that the dollar loss on the stock is reduced by the option premium received and the dollar gains on the stock are capped by the strike price (if the written call is out-of-the-money when the position is initiated). Thus, *ex ante*, compared to the buy-and-hold strategy, the covered call strategy has lower return potential and less

volatility. This *ex ante* comparative performance of covered call strategy was confirmed by Merton et al's (1978) simulation results for two different stock samples over the July 1963 to December 1975 period. We include the Merton et al's (passive) covered call strategy as it provides a benchmark for our active strategies involving stock plus options. Merton et al does not provide any objective guidance for the selection of the written call's strike price. On the other hand, in active strategies to be discussed later in this paper, the hypothetical investor selects a specific moneyness or strike price for written call options based on the open-interest based stock price predictor.

For the Merton et al type *passive* covered call strategy, on a trade initiation day our hypothetical investor buys approximately \$10,000 worth of each stock and receives premium from selling or shorting call options on the shares that expire at the end of the option month. Investor is restricted to write call options all of the same moneyness defined as percentage deviation from the initial stock price. These can be either all out-of-the-money, all at-the-money (or nearest-to-the-money), or all in-the-money call options. Since the observed strike prices are set at discrete intervals, we choose strike prices for the individual stocks to maintain roughly the same level of moneyness.

The dollar returns for the at-the-money covered call strategy is given by the following equation:

$$\Pi_{CC} = [(S_T - S_t) \times N_s + ((C_{it} \times N_c) \times 100 - T_C) - (\max\{0, (S_T - X_i) \times N_c\} \times 100)] \quad (3)$$

where, C_{it} is the closing price of the call option with strike X_i on the trade initiation day and T_C is the total transaction cost for the call option transactions, and $N_s/100 = N_c$ is the number of call option contracts written and is equal to $\$10,000/S_t$ rounded to the nearest integer. Since option trades incur a fixed ordering cost plus a per contract transaction cost, T_C can be significant percentage wise for orders of small value.

The initial investment required on a fully covered position, I_{CC} , is given by:

$$I_{CC} = N_s \times S_t - (C_{it} \times N_c) * 100 \quad (4)$$

The percentage return on investment (PROICC) is estimated as follows:

$$PROICC = (\Pi_{CC} \div I_{CC}) \times 100 \quad (5)$$

Three different equally weighted portfolios are formed using the covered call positions of the thirty individual stocks. These portfolios are named OMP (out-of-the-money calls written), AMP (at-the-money or nearest-to-the-money calls written) and IMP (in-the-money calls written). The portfolio return is just the average return on the thirty individual stocks' covered call positions.

C. Open Interest Based Active Strategy (Stock Only)

At the close of a trade initiation day, t , our active investor estimates the expected terminal stock price, S_t^Z , based on the distribution of open interests of call and put options of various strike prices but all maturing at the same date, T . If the predictor S_t^Z is greater (lower) than the contemporary security price S_t , the investor considers this as a buy (sell) signal and goes long (short) on the stock.

The expected dollar returns and the percentage returns for the active *long* stock strategy are as in equations 1 and 2. The expected dollar and percentage returns for the active *short* stock strategy are given by the following equations:

$$\Pi_{AS} = [S_{ti} - S_{Ti}] \times N_S - I \quad (6)$$

$$PROIAS = (\Pi_{AS} \div I) \times 100 \quad (7)$$

We are assuming that whether the stock is bought or sold short, the investor has to deposit the equivalent of trade value with the broker. In our case, this amount is \$10,000, i.e., the assumed investment sum. Thus, our active stock investment strategy, like the buy and hold strategy, assumes no margin buying of the stock and 100% margin requirement for short selling.

When we present the empirical results, we shall mostly rely on the average percentage return on the active strategy with respect to the thirty individual stocks in our sample. This means that the average percentage return will reflect the returns from active *long* positions in some stocks and active

short positions in others; *long* or *short* is, of course, determined individually for each stock by the comparison of the actual stock price S_t and the open interest based price predictor S_t^Z .

D. Open Interest Based Active Strategy (Stock Plus Options)

The open interest based active stock strategy can be quite risky for individual stocks. We, therefore, consider limited risk active strategies involving stocks and options. As in the active stock strategy, on each trade initiation day t , our investor compares the actual closing stock price S_t and the open interest based price predictor S_t^Z . Here, however, the investor considers the magnitude as well as the direction of predicted price movement. Suppose X is the next available strike price in the direction of the predicted price movement. It is considered a *major* price movement if S_t^Z either goes past X or at least is closer to X than it is to the current stock price S_t . Otherwise, it is categorized as a *minor* price movement. By and large, the cases identified as *major* (*minor*) price movements in our sample represented a predicted change of *more* (*less*) than 5% (2%) from the current stock price.

According to the direction and the magnitude of the predicted stock price movement, we have the following four cases:

ASP1: The price predictor signals a minor *upward* movement in the stock.

ASP2: The price predictor signals a major *upward* movement in the stock.

ASP3: The price predictor signals a minor *downward* movement in the stock.

ASP4: The price predictor signals a major *downward* movement in the stock.

We shall now discuss the four cases above and the active trading strategies our investor will follow under these cases.

D.1 ASP1: The Price Predictor Signals a Minor Upward Movement in the Stock

If the price predictor indicates a minor *upside* for a stock, our investor buys the stock at the closing market price on the trade initiation day and writes a call option. The strike price of the written call option is chosen based on the upside potential indicated by the price predictor and the availability of strike prices. If, for example, the current stock price is \$55, the predicted stock price at

maturity is \$57 and the available strike price that is immediately higher than \$57 is \$60, we consider this as a *minor* upside from the current stock price of \$55. Accordingly, our investor writes a covered call at the strike price of \$60.

The dollar returns for the strategy ASP1 is given by the following equation:

$$\Pi_{ASP1} = [(S_T - S_t) \times N_S + (C_{xi} \times N_C - T_C) \times 100 - \max\{0, (S_T - X_i) \times N_C\} \times 100] - I_{ASP1} \quad (8)$$

where $X_i \cong S_t^Z$

Here the symbol “ $\cong$ ” means closest to S_t^Z in the direction of the predicted price movement.

The initial investment required for the strategy ASP1, is given by:

$$I_{ASP1} = N_S \times S_t - (C_{xi} \times N_C) \times 100 \quad (9)$$

And the percentage return on investment (PROIASP1) is estimated as follows:

$$PROIASP1 = (\Pi_{ASP1} \div I_{ASP1}) \times 100 \quad (10)$$

In the equations above, $N_S = N_C \times 100$, is the number of shares that can be purchased with \$10,000 rounded to the nearest 100. This is because in order to write one fully covered call option contract, the investor needs to have 100 shares. For the sake of convenience, we allow the investor to buy 200 shares and write 2 call contracts if the number of shares that could be purchased with \$10,000 is 180. Therefore, the actual net investment could be higher or lower than \$10,000. The percentage return, PROIASP1, is calculated on the net investments I_{ASP1} . The options transaction cost, T_C , is the transaction cost of writing N_C option contracts. Lastly, C_{xi} is the time t closing price of the call option with strike X_i .

If the stock does move *up* but does so substantially (not expected), the return from the strategy ASP1 will be positive but less than the active stock strategy AS (*long*). However, if the stock stays roughly the same, moves *up* but not beyond the chosen strike for the written call, or in fact *drops* a little, then the strategy ASP1 may still provide a positive return that is greater than the return from the strategy AS (*long*). If the stock moves *down* by more than the net proceeds from the written call, the return from ASP1 will be negative but it will not be as bad as the return from the strategy

AS (*long*). Overall, the strategy ASP1 provides better return prospect than the strategy AS (*long*) except when the stock makes a major *upward* move unexpectedly.

The difference between the strategy ASP1 and Merton et al style covered call strategy is that the latter ignores potential information contained in the distribution of open interests. That is, ASP1 is an *active* counterpart of the Merton et al style *passive* covered call strategy. If the investor follows Merton et al style, then only by chance the investor will choose to write the out-of-the-money call option with a strike close to the predicted stock price at maturity. To the extent the distribution of open interest does contain some information and the investor chooses to write out-of-the-money call options for all thirty stocks, the *passive* covered call portfolio return is expected to be lower than the ASP1 return. The variability of passive covered call portfolio return will be higher than that of ASP1 if the distribution of open interest is highly informative. Therefore, like any other comparison of active versus passive trading, the precision of information (contained in the open interests) is an important ingredient in determining the relative risk-return tradeoff between the active versus passive limited risk strategies of covered call writing.

In the empirical section later in this paper, we report the cross-sectional (across the sample stocks) average as well as the standard deviation of the various strategies. Based on these *ex-post* measures, if we find a better risk-return tradeoff for an *active* strategy compared to a *passive* strategy, then we can feel more confident about the *ex ante* superiority of the *active* strategy.

D.2 ASP2: The Price Predictor Signals a Major Upward Movement in the Stock

If the price predictor indicates a major *upside* for a stock, as in strategy ASP1 our investor pursues a limited risk active covered call strategy. The investor buys the stock at the closing market price on the trade initiation day and sells a deep-out-of-the-money (significantly higher strike) call option; the strike is chosen based on the upside potential indicated by the price predictor and the availability of strike prices. If, for example, the predicted stock price at maturity is \$64, signaling a major upside from the current stock price of \$55, our investor writes a covered call at strike around 65. If, on the other hand, the predicted stock price at maturity were \$61 or even \$59, still signaling a major upside according to our classification, the chosen strike would have been \$60.

The dollar return, the investment and the percentage return for the strategy ASP2 are then as in equations (8), (9) and (10). This, however, will only make sense if the premium received for the deep-out-of-the-money calls written exceeds the options transaction cost. Since the options in our

sample have less than one month to expiration, when the predictor signals a major upward movement, we find some instances where the closing option price of the prescribed strike is either too low or the contract had no trade on the initiation day t . In these circumstances, our hypothetical investor pursues a somewhat different bullish strategy. The investor now buys shares and at-the-money options costing a total of \$10,000. Thus, here we have:

$$N_S @ 100 \times N_C @ \$10,000 / [S_t + C_{yi}] \quad (11)$$

where C_{yi} is the time t closing price of a call option with strike $X_{yi} @ S_t$. Since the premium for the deep-out-of-the-money ($X_{yyi} @ S_t^Z \gg S_t$) call option net of the transaction cost (when jointly ordered with the purchase of at-the-money call option) will be of relatively small magnitude, we continue to include the writing of the deep-out-of-the-money call option. The dollar return, the initial investment and the percentage return on ASP2 are then estimated as:

$$\Pi_{ASP2} = \left[(S_T - S_t) \times N_S + \max\{(S_T - X_{yi}) \times N_C, 0\} \times 100 - ((C_{yi} \times N_C) \times 100 + T_C) \right] \\ + ((C_{yyi} \times N_C) \times 100 - T_C) - \max\{(S_T - X_{yyi}) \times N_C, 0\} \times 100 \quad (12)$$

$$X_{yi} \cong S_t, \quad X_{yyi} \approx S_t^Z$$

$$I_{ASP2} = N_S \times S_t + (C_{yi} \times N_C) \times 100 - (C_{yyi} \times N_C) \times 100 \quad (13)$$

$$PROI_{ASP2} = (\Pi_{ASP2} \div I_{ASP2}) \times 100 \quad (14)$$

The purchase of at-the-money call options to partially replace the purchase of shares makes the strategy ASP2 more bullish (more leveraged) than when deep-out-of-the-money option premium exceeds the options transaction cost. However, the number of such cases is rather small and has no material impact on the overall (portfolio) returns of the strategy ASP2. Hence, for ASP2, we shall pretend from hereon that the option premium received is large enough to handily offset the transaction cost.

If the stock does move *up* as expected, the return from the strategy ASP2 will be positive and greater than both the active stock strategy AS (*long*) and the active covered call strategy ASP1

(where somewhat out-of-the-money call option is written). If the stock stays roughly the same, does not make a major upward move, or *drops* a little, then the strategy ASP2 may still provide a positive return that is greater than the return from the strategy AS (*long*). Although, under such circumstances, ASP1 would have yielded a greater return than ASP2 since the call premium received under ASP1 would have been higher. If the stock moves *down* by more than the net proceeds from the written call, the return from ASP2 will be negative but it will not be as bad as the return from the strategy AS (*long*). Overall, the strategy ASP2 provides better return prospect than the strategy AS (*long*) except when the stock makes a major *upward* move beyond the predicted price. In that case, AS (*long*) would outperform ASP2.

D.3 ASP3: The Price Predictor Signals a Minor Downward Movement in the Stock

If the price predictor indicates a minor *downside* for a stock, our investor writes a covered call at a strike near the closing stock price on the trade initiation day. If the stock goes down to the somewhat lower predicted level but not below, the loss on the *long* stock will be outweighed by premium received for the at-the-money call written assuming that the call premium is greater than the expected drop in the stock. In case the stock stays flat or moves up, there is no loss on the *long* stock as the investor either keeps the stock or gives it away at the strike close to the purchase price. Meantime, the investor bags the call premium. The worst case for this strategy is when the stock drops significantly below the predicted level. In that case, the strategy ASP3 will lose money although the call premium cushions the loss to a degree.

If the investor instead went for short selling, the dollar gains would be more if the stock drops as predicted. However, not only the required investment is larger (due to the premium received under ASP3) relative to the strategy ASP3, the short selling strategy is destined to lose money if the stock moves up instead of moving down. The same is true for naked put buying. In other words, short selling and naked put buying are more volatile bearish alternatives than the strategy ASP3.

The dollar returns, the initial investment and the percentage return for the strategy ASP4 are given by the following equations:

$$\Pi_{ASP3} = \left[(S_T - S_t) \times N_S - (\max \left\{ (X_{vi} - S_T) \times N_C, 0 \right\} \times 100) + (C_{vi} \times N_C) \times 100 - T_C \right] \quad (15)$$

$X_{vi} = \text{call strike closest to } S_t.$

$$I_{ASP3} = N_S \times S_t - (C_{vi} \times N_C) \times 100 \quad (16)$$

$$PROIASP3 = (\Pi_{ASP3} \div I_{ASP3}) \times 100 \quad (17)$$

In the above equations, P_{zi} is the price of the put option with strike X_{zi} and C_{vi} is the price of the call contract with strike X_{vi} . The transaction costs are T_C for N_C call option contracts written and $N_C = N_S/100$ where N_S is $\$10,000/S_t$ rounded to the nearest 100.

D.4 ASP4: The Price Predictor Signals a Major Downward Movement in the Stock

If the price predictor indicates a major *downside* for a stock, our investor writes a covered call at a strike close to the predicted stock price at maturity and buys a put option with the strike near or above the closing stock price on the trade initiation day. If the stock goes down to the predicted level or below, the loss from the long stock will be covered by the put option's payoff, the written call be worthless, and the net premium received (price of the in-the-money call written minus the price of the at-the-money put option bought) can be considered as profit. The net premium plus the put payoff would likely exceed any loss from the exercise of written call when the stock does not drop all the way to the predicted low price. In case the stock moves up, profit from the stock would offset further losses on the written call. The worst case for this strategy if the stock remains flat and the net premium is less than the intrinsic value of the written in-the-money call.

If the investor instead went for short selling, the dollar gains would be more if the stock does actually drop significantly as predicted. However, not only the required investment is larger (when the net premium is positive), the loss is also greater if the stock moves up instead of moving down. The same is true for naked put buying. In other words, short selling and naked put buying are more volatile bearish alternatives than the strategy ASP4.

The dollar returns, the initial investment and the percentage return for the strategy ASP4 are given by the following equations:

$$\Pi_{ASP4} = \left[\begin{aligned} &(S_T - S_t) \times N_S - (P_{Zi} \times N_P) \times 100 + T_P + (\max\{(X_{Zi} - S_T) \times N_P, 0\} \times 100) \\ &- (\max\{(X_{Vi} - S_T) \times N_C, 0\} \times 100) + (C_{Vi} \times N_C) \times 100 - T_C \end{aligned} \right] \quad (18)$$

X_{Zi} = put strike closest to S_t , and X_{Vi} = call strike closest to S_t^Z .

$$I_{ASP4} = N_S \times S_t - (C_{Vi} \times N_C) \times 100 + (P_{Zi} \times N_P) \times 100 \quad (19)$$

$$PROI_{ASP4} = (\Pi_{ASP4} \div I_{ASP4}) \times 100 \quad (20)$$

In the above equations, P_{Zi} is the price of the put option with strike X_{Zi} and C_{Vi} is the price of the call contract with strike X_{Vi} . The transaction costs are T_C for N_C call option contracts written and T_C for N_P put option contracts bought, and $N_C = N_P = N_S/100$ where N_S is $\$10,000/S_t$ rounded to the nearest 100.

IV. Evidence on Comparative Performance

In this section, we *first* discuss (Table III) how the active strategy performs for a given option month and trade initiation day and compare this with the naïve investor's buy and hold strategy. *Second*, the prediction accuracy of the open interest based price predictor is analyzed (Table IV). *Third*, the comparative performance results for the various option months and trade initiation days are reported for the various passive and active strategies (Table V). These strategies are the naïve investor's buy and hold strategy (NI), the S&P 500, the Merton et al (1978) style passive covered call strategy (OMP: out-of-the-money call written, AMP: at-the-money call written and IMP: in-the-money call written), the open interest based stock only limited risk active strategy (AS) and the stock plus options limited risk active strategy (ASP, ASP1: minor up predicted, ASP2: major up predicted, ASP3: minor down predicted, and ASP4: major down predicted). For the active strategies, we only report the performance results for the CN (Consider News) situation where the stocks with major

news events are excluded from the portfolio. The performance figures for a strategy that we report are the cross-sectional (across the sample stocks) average and standard deviation of holding period Percentage Return on Investment (PROI). However, for the most part, our discussion focuses on the average return. For the sake of convenience, we shall often use the term mean return or simply return when referring to the average PROI.

A. How Does the Active Strategy Perform?: An Example

To obtain a sense of how the active positions are established and aggregated to arrive at portfolio results, Table III provides an example of detailed stock by stock prediction, strategy and performance for the stock only active strategy (AS). It shows for the February option month of 1999 the comparative performances of the buy and hold strategy (NI or Naïve Investor Portfolio), the stock only active strategy considering news (CN or OPP Portfolio) and the stock only active strategy ignoring news (IN Portfolio). The active strategies here were initiated on 2F or the second Friday of the February option month (the third last Friday counting backward from the option expiration Friday in February 1999). That is, the options had about 14 days to expiration as of the trade initiation day.

The first column of Table III indicates the symbols of the sample stocks. The second column displays the closing prices, S_T , at the option maturity date. The third column shows the open interest based stock price prediction, S_t^Z , where t is the second Friday of the February 1999 option month. The fourth column indicates the closing stock price, S_t , on our trade initiation day t . The fifth column shows the direction of movement of the stock signaled or predicted by the predictor. If $S_t > S_t^Z$, it is identified as “Down”. If, on the other hand, If $S_t < S_t^Z$, it is identified as "Up". The sixth column shows the stock only active trading strategy (BUY for *long* and SS for *short selling*) for our hypothetical learner investor. The seventh column refers to the number of shares the investor buys or sells short with \$10,000. The last three columns display the PROI for 14 days holding period for the NI, CN (OPP) and the IN portfolios.

The results from the table indicate that our hypothetical investor could have earned a return of 0.86% for the two-week holding period in the February 1999 option month following the stock only active strategy and excluding the one stock (MO) that had a major news event during the holding period. If this wisdom or foresight is taken away, the return to the stock only active strategy would have been 0.39%. While the impact of the news event is important, it however does not

change the ranking of the active strategy vis-à-vis the naïve investor's passive strategy (NI) of blindly buying the stocks and holding them. The NI strategy/portfolio provides a negative return of – 0.80%. Thus, the active CN (IN) strategy had an incremental 1.66% (1.19%) for the two-week holding period or about 43.16% (30.94%) annualized.

While we have not presented any risk measure in Table III and the table concerns only one initiation day in one option month, it seems that the return advantage of the active strategy compared to the buy and hold strategy is quite convincing. Thus there is preliminary indication that options open interest contain information about future stock movement that can be used for profitable trading. The volatility of the active strategy returns relative to the volatility of the buy and hold returns have to be quite high for nullifying the active strategy's attractiveness.

B. How Accurate is the (Up or Down) Prediction of the Open Interest Based Predictor?

In Table III, we see that while the stock only active strategy did have a mean return advantage during the two-week period of the February 1999 option month, the up or down predictions for the sample individual stocks were not always right. If the return advantage arises due to accurate prediction in only a minority of stocks and only in a few months, then the strategy may not be reliable for replication in general. We now, therefore, look at the prediction (up or down) accuracy of the open interest based predictor in the six option months in our sample considering all four trade initiation days within an option month. Since there are thirty stocks in our sample, we have a total of 120 cases of prediction in each of the six option months.

In the three option months of February (59%), April (70%) and May (65%), a majority of the sample stocks actually lost their value during the holding periods. On the other hand, in the option months of March (53%), June (77%) and July (63%), a majority of the sample stocks actually marched higher during the holding periods. Thus, loosely speaking we might refer to February, April and May as the “down” option months in our sample and to March, June and July as the “up” option months.

In comparison, our option based predictor calls for a “down” option month in all option months except June. Thus there seems to be a downward bias in our open interest based prediction. This is somewhat expected as our prediction method is solely based on the stock-specific open interest of options and does not take into account any market wide factor, nor does it predict up or

down for a portfolio of stocks. We also ignore volume and recent changes in the open interest that might have contributed to the prediction error.

Let us now look at the percentage of cases the open interest based prediction was correct. Of the cases the prediction was “up”, it proved to be a correct call in 67% (February), 50% (March), 54% (April), 55% (May), 77% (June) and 87% (July) of the cases. Of the cases the prediction was “down”, it proved to be a correct call in 60% (February), 45% (March), 80% (April), 81% (May), 24% (June) and 56% (July) of the cases. Averaging over the six option months, the accuracy of the “up” prediction is 65% and that of the “down” prediction is 58%. Thus, the “up” predictions may appear more accurate than the “down” predictions. But in the “down” option months (February, April and May) when a majority of sample stocks suffered loss, the accuracy of “down” prediction is 60%, 80% and 81% respectively, averaging to 74% accuracy. In contrast, in the “up” option months (March, June and July) when a majority of sample stocks marched higher, the accuracy of “up” prediction is 50%, 77% and 87% respectively, averaging to 71% accuracy.

We, therefore, conclude that the prediction accuracy of the open interest based predictor is reasonably good. The overall prediction accuracy is not due to accuracy only in specific option month or trade initiation day. In other words, our evidence so far lends broad support to the information content of the options open interest.

C. Comparative Performance Results

Lastly, Table V presents the comparative performance results for the various passive and active strategies in the six option months considering the four alternative trade initiation days (2F, 2M, LF, EXM) within each option month.

Considering all 120 cases of prediction, a passive investor investing in the S&P 500 would have earned 1.53% return on average. If the passive investor followed the equally-weighted buy and hold strategy, the average return would have been 1.00%. Considering that the average holding period is about 11 to 12 days, these returns translate to about 49% annualized return for the S&P 500 and about 32% annualized return for the equally weighted portfolio of the thirty blue chip sample stocks. Given that 1999 was a stellar year for stocks, these returns appear realistic.

Now consider the aggregate performance of the open interest based stock only active strategy. Following this strategy, our hypothetical learner investor could expect to earn 9.05% return on average. This translates to an annualized return of about 2600%. By any means, the return

advantage of the open interest based active strategy seems convincing. That this return advantage did not arise due to better performance in just one or two months or for just one or two specific trade initiation days can be observed from the detailed performance numbers in Table V. Whichever stock only strategy (S&P 500, NI, AS) earned the highest return in a given option month for a given trade initiation day is highlighted in Table V. For example, the AS strategy had the highest return in 16 of the 24 option month/ trade initiation day combinations and these cases are well spread over the various months and trade initiation days.

One might wonder whether the much enhanced performance of the stock only active strategy comes at the cost of significantly higher risk or volatility. As a proxy for risk or volatility, we also present in Table V the cross-sectional standard deviation (in *italics*) of holding period return. On average, the standard deviation of the AS strategy return is 22.64% while that of the buy and hold strategy is 25.07%.¹⁷ This gives us confidence that the return advantage of the active strategy will not be washed away by a significantly higher risk.

Let us now turn to the limited risk strategies. After all, our learner investor is subject to the risk of gleaning incorrect information from the distribution of options open interest and as such may hedge a bit by undertaking limited risk directional speculation instead of naked speculation (as in the active stock only strategy above). The benchmark that we use for the limited risk strategies is the Merton et al (1978) type covered call strategy. Since this strategy provides no guidance for the selection of the strike for the written call option, it can be considered a passive limited risk strategy. In contrast, although our stock plus option strategy also uses covered call, the strike of the written call is carefully chosen according to the direction and the magnitude of stock price movement predicted by the open interest based price predictor.

In aggregate, the passive covered call strategy earns 1.96% (OMP: out-of-the-money call written), 2.43% (AMP: at-the-money call written) and 2.14% (IMP: in-the-money call written). In sharp contrast, our stock plus options limited risk active strategy (ASP) earns 11.20%. Once again, the advantage of the active strategy is not due to return advantage in particular option month or for a given trade initiation day. This advantage prevails in 21 out of the 24 option month/ trade initiation day combinations in our sample. Looking at the standard deviations, the risk of the limited risk active strategy appears consistently lower than that of the passive out-of-the-money covered call strategy. In general, the risk of the active strategy is greater compared to the risks of the passive at-

¹⁷ The standard deviation of the S&P 500 holding period return for a given option month/ trade initiation day combination cannot be calculated as it is a single number.

the-money and in-the-money covered call strategies. This is, of course, expected as our limited risk active strategies mostly involve writing out-of-the-money call options and as such their volatility is expected to be higher relative to the strategy of always writing at- or in-the-money call options.

Overall, the comparative performance results indicate an impressive trading advantage of predictions based on the distribution of options open interest. This advantage seems pervasive and does not seem to come at the cost of significantly higher risk.

V. Summary and Concluding Remarks

In this paper, we use daily closing data on CBOE options of 30 stocks during February through July of 1999 to investigate whether options open interest contains information that can be used for trading purposes. Individual stock price at option maturity is first predicted based on the distribution of options open interest. Several stock only and stock plus options directional trading strategies are then considered after comparing the predicted stock price at maturity and the actual stock price at the trade initiation date.

We find the prediction of stock price movement based on the distribution of options open interest to have reasonably good accuracy. The prediction accuracy is not due to accuracy only in specific option month or trade initiation day. In our sample, the open interest based active trading strategies generate better returns compared to the passive benchmarks. The stock only active strategy yields significantly higher return than the S&P 500 and the naïve investor's buy and hold strategy involving the sample stocks. Since our hypothetical learner investor faces the risk of incorrect information or inaccurate learning, the investor might prefer limited risk speculative strategies involving stock plus options. In this context, our benchmark is Merton et al (1978) style covered call strategy. Here also, we find that the open interest based active strategy provides significantly higher return than the passive covered call strategies.

Not only the risk of the active strategies (naked and limited risk) seems close to the risk of the benchmarks, the magnitude of the return advantage seems too high to be nullified by any risk disadvantage there may be due to the use of a rough proxy for risk in this paper. We, therefore, conclude that the equity options open interest contains valuable information that is attractive for trading purposes.

Our evidence has important implications for researchers and practitioners. The information content of derivative market activity that we find in this paper lends support to a growing theoretical and empirical literature. Many (e.g., Copeland and Galai (1983), Glosten and Milgrom (1985), Kyle (1985), Easley and O'Hara (1987), He and Wang (1995), Easley et al (1998), John, Koticha, Narayanan, and Subrahmanyam (2000)) in this literature suggest that the derivatives such as options cannot be considered non-redundant in the context of asymmetric and differential information or differential interpretation of information. In fact, as suggested by Cherian and Vila (1997), Cherian (1998), and Cherian and Weng (1998), informed traders in possession of volatility related information can only use the options market and this may have implication for the information content of the option prices and options market activity.

The evidence in this paper concerns directional trading only and have ignored volatility trading. Given that even these simpler trading strategies seem to indicate high information content of derivatives market activity, it is expected that the information content will be even greater once more complex strategies based on the prediction of direction as well as volatility are entertained. For example, one may use the distribution of open interests to forecast the cumulative volatility to expiration and then compare with the volatility implied by the Black-Scholes model or generated with other standard volatility models.¹⁸ The implied volatility as well as most standard volatility models are based on option prices and/or the underlying asset price series. If information impacts neither the option price nor the current level of the underlying asset price and the informed traders do use the options market to trade on their information, then only the derivatives market activity will contain that information.

This paper shows one clear advantage of using derivatives market activity instead of prices to imply information. The information implied from the derivative prices is about the risk-neutral distribution of the underlying asset. While it is certainly useful in many applications, in many other applications it is the physical distribution of the asset that is of most interest. For example, in stock only portfolio optimization, the moments needed are those from the physical distribution. Similarly, practitioners interested in estimating the beta and the cost of capital will find information about the physical distribution more useful. In these and other applications where the physical distribution is of interest, the derivatives market activity measures may be quite helpful to the extent these activities

¹⁸ Alternatively, one may combine the distribution of open interests, the distribution of option volume, historical time series based distribution, the risk neutral distribution and/or the implied volatility surface to minimize forecast error of the predicted price or higher order moments of the asset's true future distribution.

do contain information about the physical distribution. It is with respect to this latter regard, this paper provides hope for future applications.

Lastly, it is to be mentioned that we considered only US equity options and only one measure of activity in this market, namely the open interest. It remains to be seen how the approach will work in the context of options on other assets (such as foreign exchange, fixed income, and commodity), derivatives of other types (such as futures and futures options) and derivatives traded in other countries.

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Table I
Sample Firms

The sample consists of the 30 firms representing various sectors with active CBOE options trading during February through July of 1999. For each stock, the table gives the stock trading ticker symbol, the sector it represents on the CBOE and the major index in which it was included at the time.

Symbol	Name of the company	Sector	Index
CTL	Century Telecommunications, Inc	Services (Communication)	S&P 500
C	Citigroup Inc.	Services (Bank)	DJIA
DD	E.I. Dupont De Nemours	Basic Materials	DJIA
GE	General Electric Company	Conglomerates	DJIA
SUNW	Sun Micro Systems Inc.	Technology (HW)	NASDAQ 100
KO	Coca-Cola Company	Consumer (NC)	DJIA
PG	Proctor and Gamble Co.	Consumer (NC)	DJIA
WMT	Wall-Mart Stores, Inc.	Services (Retail)	DJIA
ABX	Barrick Gold Corp.	Basic Material	S&P 500
AMAT	Applied Materials	Technology (Semi Eq)	NASDAQ 100
AMGN	Amgen, Inc.	Healthcare (Drugs)	NASDAQ 100
AMR	American Airlines, Inc.	Services (Transport)	DJIA
ASND	Ascend Communications	Technology (Network)	NASDAQ Comp.
CMB	Chase Manhattan Corp.	Services (Bank)	S&P 500
COMS	3Com Corporation	Technology (Network)	NASDAQ Comp.
CPQ	Compaq Computer Corp.	Technology (HW)	S&P 500
DAL	Delta Airlines, Inc.	Services (Transport)	DJIA
DIS	Walt Disney Company	Services (Recreation)	DJIA
UTX	United Technologies	Conglomerates	DJIA
CSCO	Cisco Systems, Inc.	Technology (Network)	NASDAQ 100
WCOM	MCI Worldcom, Inc.	Services (Comm.)	NASDAQ 100
DELL	Dell Computer Corporation	Technology (HW)	NASDAQ 100
SLB	Schlumberger Limited	Energy (Oil & Equip)	S&P 500
ORCL	Oracle Corporation	Technology (Software)	NASDAQ 100
MU	Micron Technology, Inc.	Technology (Semicon)	S&P 500
MOT	Motorola, Inc.	Technology (Comm.E)	S&P 500
MO	Philip Morris Companies	Consumer (NC Tob)	DJIA
ALTR	Altera Corp.	Technology (Semi. EQ)	NASDAQ Comp
MCD	McDonald's Corporation	Services (Food)	DJIA
S	Sears, Inc.	Services (Retail)	DJIA

Data Source: Yahoo. Market Guide, March 17, 1999

Table II
Sample Firms and Market Statistics

The table gives market beta , market capitalization, shares floating, shares outstanding and average daily volumes of trade for each selected firm.

Symbol	Beta	Market Capitalization (Billions)	Shares Floating (Millions)	Shares Outstanding (Millions)	Daily Volume (Millions)
CTL	0.96	6.72	60.00	92.40	0.31
C	1.41	146.10	2190.00	2260.00	9.11
DD	0.70	64.00	1010.00	1300.00	2.99
GE	1.14	359.40	3200.00	3270.00	4.58
SUNW	1.42	43.90	373.70	385.30	7.68
KO	1.02	168.30	2000.00	2470.00	3.72
PG	0.93	123.10	1310.00	1330.00	2.19
WMT	0.88	218.30	1180.00	2220.00	3.10
ABX	1.12	6.84	300.80	376.00	1.31
AMAT	2.59	23.80	369.20	373.00	8.82
AMGN	1.16	39.70	498.80	509.00	4.78
AMR	1.22	11.30	169.60	182.30	1.64
ASND	1.62	17.00	190.90	216.90	2.41
CMB	1.41	70.80	835.80	844.20	4.08
COMS	1.78	8.95	340.00	358.80	9.45
CPQ	1.36	54.20	1670.00	1700.00	16.00
DAL	0.87	10.00	102.00	141.60	1.17
DIS	0.93	72.90	2010.00	2060.00	6.21
UTX	1.26	29.70	184.60	225.10	0.71
CSCO	1.35	170.70	1570.00	1600.00	15.50
WCOM	1.48	171.90	1710.00	1830.00	11.60
DELL	1.70	107.50	1960.00	2540.00	44.20
SLB	1.05	32.30	519.00	546.40	3.26
ORCL	1.45	42.70	1090.00	1440.00	15.60
MU	2.39	12.90	101.50	247.70	5.13
MOT	1.20	43.60	576.20	600.20	3.04
MO	0.73	101.90	2230.00	2430.00	7.24
ALTR	1.99	5.93	80.80	97.30	2.44
MCD	0.96	59.90	1340.00	1360.00	3.78
S	1.05	341.30	341.30	383.50	1.43

Source of Data: Yahoo. Market Guide, March 17, 1999.

Table III**Individual Stock Performance of Buy and Hold and Stock Only Active Strategy (AS)**

For this table, the trade initiation day (t) is the second Friday of the February 1999 option month that runs from the first day of trading after option expiration in January 1999 to the last day of trading (T) before option expiration day in February 1999. Time to expiration or the holding period here is 14 calendar days. S_t and S_T are the actual closing prices of the stock at t and T. S_t^Z is the open interest based prediction at t for S_T . It is the weighted average of the strike prices that had open interest at t in call and/or put options maturing at T. The weight for a given strike price is the combined open interest of call and put options (maturing at T) at that strike price relative to the aggregate open interest of call and put options (maturing at T) of all strike prices.

Stock Symbol	S_T	S_t^Z	S_t	Signal	Active Strategy	N_S	PROI (Percentage Return on Investment)		
							Portfolio Type:		
							IN	CN	NI
							(OPP)		
ABX	18.44	20.21	20.25	DOWN	SS	494.00	8.91	8.91	-8.91
ALTR	58.50	61.85	56.75	UP	BUY	176.00	2.96	2.96	2.96
AMAT	68.69	53.42	60.69	DOWN	SS	165.00	-13.34	-13.34	13.34
AMGN	124.12	112.20	119.63	DOWN	SS	84.00	-4.26	-4.26	4.26
AMR	53.75	61.25	59.94	UP	BUY	167.00	-10.24	-10.24	10.24
ASND	77.18	77.18	77.00	UP	BUY	130.00	0.33	0.33	0.33
C	54.18	53.39	52.88	UP	BUY	189.00	2.40	2.40	2.40
CMB	76.06	68.47	76.43	DOWN	SS	131.00	0.36	0.36	-0.36
COMS	33.18	47.24	34.19	UP	BUY	293.00	-2.78	-2.78	-2.78
CPQ	41.12	46.04	43.63	UP	BUY	229.00	-5.84	-5.84	-5.84
CSCO	97.12	102.95	101.25	UP	BUY	99.00	-3.85	-3.85	-3.85
CTL	62.32	66.42	67.38	DOWN	SS	149.00	7.14	7.14	-7.14
DAL	55.06	54.20	57.56	DOWN	SS	174.00	4.20	4.20	-4.20
DD	52.75	55.65	55.12	UP	BUY	182.00	-4.00	-4.00	-4.00
DELL	80.12	76.46	100.44	DOWN	SS	100.00	19.88	19.88	-19.88
DIS	34.12	34.28	34.25	UP	BUY	292.00	-0.37	-0.37	-0.37
GE	100.38	99.26	98.00	UP	BUY	102.00	2.39	2.39	2.39
KO	65.75	67.12	62.06	UP	BUY	161.00	5.86	5.86	5.86
MCD	85.56	78.74	80.31	DOWN	SS	125.00	-6.95	-6.95	6.95
MO	39.94	50.07	46.13	UP*	BUY	217.00	-13.33	0.00	-13.33
MOT	67.38	67.58	66.44	UP	BUY	151.00	1.74	1.74	1.74
MU	64.12	64.47	70.25	DOWN	SS	143.00	8.31	8.31	-8.31
ORCL	54.19	50.76	56.19	DOWN	SS	178.00	3.54	3.54	-3.54
PG	91.82	89.38	84.88	UP	BUY	118.00	8.35	8.35	8.35
S	39.69	39.85	40.31	DOWN	SS	248.00	1.57	1.57	-1.57
SLB	49.19	50.42	53.00	DOWN	SS	189.00	7.03	7.03	-7.03
SUNW	96.94	100.19	100.63	DOWN	SS	100.00	3.06	3.06	-3.06
UTX	125.44	104.88	125.00	DOWN	SS	84.00	-0.35	-0.35	0.35
WCOM	84.18	73.32	76.25	DOWN	SS	131.00	-10.28	-10.28	10.28
WMT	84.75	80.92	84.25	DOWN	SS	119.00	-0.85	-0.85	0.85
Average PROI (Percentage Return on Investment)							0.39	0.86	-0.80

*A major news event took place for the stock after the trade initiation day.

Table III--Continued

If $S_t < S_t^Z$, then the prediction or signal is an upward move (UP) for the stock from t to T and the hypothetical active investor goes long on the stock (BUY). If $S_t > S_t^Z$, then the prediction is a down move (DOWN) for the stock from t to T and the hypothetical active investor goes short on the stock (SS). For each stock, the number of shares (N_S) bought or shorted is $\$10,000 / S_t$. We assume 100% margin deposit for both stock purchase and short selling. Stock transaction costs are assumed to be zero. The percentage return on investment (PROI) is calculated as $100 \times \text{dollar return } (P) / \text{Investment } (I)$. For the buy and hold strategy and the stock only active strategy, I is $\$10,000$. For the buy and hold strategy and the *long* stock active strategy, $\Pi = S_T - S_t$ per share. For the *short* stock active strategy, $\Pi = S_t - S_T$ per share.

Portfolio type NI (Naïve Investor) indicates a strategy of buying on trade initiation day and holding till next option expiration day with respect to each of the thirty sample stocks. In the CN and IN portfolios, on the other hand, the hypothetical investor goes *long (short)* on the stock if the open interest based price predictor signals an *UP (DOWN)* market for the stock from the trade initiation day to the next option expiration day. The difference between CN (Consider News) and IN (Ignore News) is that IN includes active strategy positions in all stocks even if some of the stocks had major news events after trade initiation day that led to major swings in the stock price. CN, on the other hand, excludes the news event stocks and as such assumes that our directional trader had expected such news although the direction of the stock impact was not known.

Table IV**Prediction (Up or Down) Performance of the Open Interest Based Price Predictor**

This table reports the number and the percentage (in parentheses) of cases in which the sample stocks actually moved up (Actual Up) or down (Actual Down) from the four alternative trade initiation days to the option expiration day during the six option months of February through July of 1999. For example, the February 1999 option month runs from the first day of trading after option expiration in January 1999 to the last day of trading (T) before option expiration day in February 1999. Of the total 120 possible cases during this option month (4 holding periods for each of the 30 stocks), in 49 or 41% of the cases, the stocks actually moved up and in the remaining 71 or 59% of the cases the stocks actually moved down.

This table also reports the number and percentage (in parentheses) of cases in which the sample stocks were predicted by the open interest based price predictor to move up (Predicted Up) or down (Predicted Down) from the four alternative trade initiation days to the option expiration day during the six option months of February through July of 1999.

Lastly, the table reports the number of cases where the prediction of upward movement (Correct Up) or downward movement (Correct Down) proved to be correct. The percentage in parentheses here represents the number of correct up (down) predictions as a percentage of the number of up (down) predictions.

Option Month	February	March	April	May
Actual Up (% of Total)	49(41%)	64(53%)	36(30%)	42(35%)
Actual Down (%of Total)	71(59%)	56(47%)	84(70%)	78(65%)
Predicted Up (% of Total)	42(35%)	48(40%)	35(29%)	53(44%)
Predicted Down (%of Total)	78(65%)	72(60%)	85(71%)	67(56%)
Correct Up (% of Predicted Up)	28(67%)	24(50%)	19(54%)	29(55%)
Correct Down (% of Predicted Down)	47(60%)	32(45%)	68(80%)	54(81%)
	June	July		
Actual Up (% of Total)	92(77%)	76(63%)		
Actual Down (%of Total)	28(23%)	44(37%)		
Predicted Up (% of Total)	78(65%)	54(45%)		
Predicted Down (% of Total)	42(35%)	66(55%)		
Correct Up (% of Predicted Up)	60(77%)	47(87%)		
Correct Down (% of Predicted Down)	10(24%)	37(56%)		

Table V
Comparative Performance of the Passive and Active Strategies

This table reports the Percentage Holding Period Return on Investment (PROI) for the buy and hold strategy (NI), the stock only active strategy (AS), the stock plus options limited risk active strategy (ASP), and Merton et al (1978) style covered call strategies (OMP: out-of-the-money call option written, AMP: at-the-money call option written, IMP: in-the-money call option written). The PROI on S&P 500 is also reported for the corresponding holding periods. For each strategy, the first row presents the cross-sectional average and the second row (in *italics*) presents the cross-sectional standard deviation of PROI across the sample stocks. The S&P 500 figures are just the holding period returns and not averages across the component stocks. The PROI is not annualized.

The option months considered are the six consecutive option months from February 1999 to July 1999. For example, the February 1999 option month runs from the first day of trading after option expiration in January 1999 to the last day of trading (T) before option expiration day in February 1999. Within an option month, four alternative trade initiation days (start of a holding period), t , are considered. These are the second Friday of the option month (2F), the second Monday of the option month (2M), the Friday of the week preceding the option expiration week (LF) and the Monday of the option expiration week (EXM). The opening trades on t and the closing trades on T use the closing prices of stocks and options on those dates.

The stock only active strategy is based on the directional (Upward or Downward) signal about stock price movement from t to T. The signal is generated from a comparison of S_t and S_t^Z , where S_t^Z is the open interest based prediction at t for S_T . It is the weighted average of the strike prices that had open interest at t in call and/or put options maturing at T. The weight for a given strike price is the combined open interest of call and put options (maturing at T) at that strike price relative to the aggregate open interest of call and put options (maturing at T) of all strike prices.

		Portfolio/Strategy			Portfolio/Strategy			
Option	Trade	AS	NI	S&P	ASP	Merton et al (1978) Covered Call:		
Month	Initiation	(CN)		500	(CN)	OMP	AMP	IMP
	Day, t							
FEBRUARY	2F	8.03 <i>17.18</i>	-8.06 <i>18.31</i>	-0.02	9.46 <i>12.34</i>	-1.95 <i>13.59</i>	0.64 <i>9.73</i>	2.07 <i>4.47</i>
FEBRUARY	2M	2.01 <i>17.34</i>	-1.87 <i>17.79</i>	-0.86	6.14 <i>16.13</i>	-4.84 <i>26.74</i>	5.11 <i>14.39</i>	2.38 <i>6.83</i>
FEBRUARY	LF	9.87 <i>25.44</i>	-7.70 <i>27.96</i>	2.56	7.74 <i>14.87</i>	-4.80 <i>21.47</i>	-2.27 <i>11.98</i>	2.59 <i>7.34</i>
FEBRUARY	EXM	1.79 <i>24.02</i>	4.09 <i>23.80</i>	-1.18	4.65 <i>17.70</i>	6.22 <i>24.25</i>	4.47 <i>15.01</i>	3.13 <i>10.14</i>
MARCH	2F	9.40 <i>21.22</i>	7.01 <i>23.93</i>	3.86	9.42 <i>14.49</i>	1.42 <i>15.13</i>	0.13 <i>11.10</i>	-0.98 <i>7.87</i>
MARCH	2M	-0.39 <i>15.42</i>	1.96 <i>20.96</i>	3.33	5.06 <i>15.11</i>	6.22 <i>19.42</i>	3.42 <i>11.95</i>	-0.26 <i>10.09</i>
MARCH	LF	-3.53 <i>14.88</i>	-0.04 <i>17.14</i>	1.40	5.31 <i>14.53</i>	1.32 <i>20.39</i>	3.50 <i>12.87</i>	2.57 <i>7.35</i>
MARCH	EXM	0.99 <i>21.70</i>	-5.08 <i>22.82</i>	-3.78	-0.12 <i>17.48</i>	-1.41 <i>22.75</i>	-0.18 <i>18.22</i>	-1.37 <i>15.55</i>

Table V— Continued

Option	Trade	Portfolio/Strategy			Portfolio/Strategy			
		AS	NI	S&P	ASP	Merton et al (1978) Covered Call:		
Month	Initiation	(CN)		500	(CN)	OMP	AMP	IMP
	Day, t							
APRIL	2F	5.99 20.65	-2.40 27.63	4.03	9.94 15.23	-3.98 21.18	-1.27 13.02	1.78 6.42
APRIL	2M	15.84 18.80	-7.10 22.08	-0.41	11.91 20.76	-4.07 28.36	-1.59 20.89	1.59 6.93
APRIL	LF	12.09 33.00	-14.07 40.07	-8.45	12.32 22.79	-13.47 31.47	-5.44 15.56	2.00 8.24
APRIL	EXM	18.23 41.34	-15.00 43.92	-18.10	16.31 28.38	-14.42 45.94	-11.47 24.59	1.95 10.47
MAY	2F	3.58 15.35	-1.03 15.90	-2.25	4.63 9.54	-3.27 15.92	1.73 10.26	1.57 5.19
MAY	2M	7.78 12.72	-7.31 13.96	-1.94	8.68 12.27	2.97 13.89	7.85 9.78	6.53 5.87
MAY	LF	6.51 19.28	-6.24 19.45	-2.13	8.95 16.24	-3.03 20.19	3.95 13.37	3.69 5.36
MAY	EXM	18.10 35.40	-14.20 37.02	-4.28	19.78 31.92	-7.98 26.47	-0.56 20.27	1.67 7.50
JUNE	2F	13.20 22.61	12.61 23.51	2.34	12.66 17.91	9.80 19.49	6.08 11.84	3.08 5.95
JUNE	2M	11.11 20.78	14.11 21.48	1.60	14.42 16.74	8.97 13.81	5.13 7.62	1.73 3.49
JUNE	LF	13.29 34.07	20.34 39.00	14.73	18.76 21.90	14.67 19.96	7.86 12.64	2.90 9.33
JUNE	EXM	16.99 37.26	27.28 43.92	23.37	29.70 37.74	21.47 30.36	9.56 19.58	3.10 12.10

Table V— Continued

Option	Trade	Portfolio/Strategy			Portfolio/Strategy			
		AS	NI	S&P	ASP	Merton et al (1978) Covered Call:		
Month	Initiation	(CN)		500	(CN)	OMP	AMP	IMP
	Day, t							
JULY	2F	9.71 23.36	8.47 24.29	4.09	11.16 12.44	7.85 14.57	4.28 9.89	1.58 5.03
JULY	2M	12.55 17.05	8.44 18.52	5.71	13.48 11.46	8.97 16.63	5.98 10.97	2.34 6.02
JULY	LF	8.71 15.15	3.02 17.72	4.26	10.69 11.03	6.46 21.36	3.87 13.89	1.95 7.13
JULY	EXM	15.56 19.23	6.75 20.38	8.74	17.74 15.75	13.97 36.26	7.57 24.68	3.78 11.14
Aggregate:								
Average PROI		9.05	1.00	1.53	11.20	1.96	2.43	2.14
Average St. Dev.		22.64	25.07		19.05	22.48	14.34	7.74

If $S_t < S_t^Z$, then the prediction or signal is an upward move (UP) for the stock from t to T and the hypothetical active investor goes long on the stock (BUY). If $S_t > S_t^Z$, then the prediction is a down move (DOWN) for the stock from t to T and the hypothetical active investor goes short on the stock (SS). For each stock, the number of shares (N_s) bought or shorted is $\$10,000 / S_t$. We assume 100% margin deposit for both stock purchase and short selling. Stock transaction costs are assumed to be zero. The percentage return on investment (PROI) is calculated as $100 \times \text{dollar return } (P) / \text{Investment } (I)$. For the buy and hold strategy and the stock only active strategy, I is \$10,000. For the buy and hold strategy and the *long* stock active strategy, $\Pi = S_T - S_t$ per share. For the *short* stock active strategy, $\Pi = S_t - S_T$ per share. The stock plus options limited risk active strategy (ASP) is based on the direction (Upward or Downward) as well as the magnitude of stock price movement as predicted by the open interest based predictor S_t^Z . The magnitude is considered a *major* move if S_t^Z either goes past or at least is closer to the next available strike in the direction of the price move than it is to the current stock price S_t . By and large, this meant a change of more than 5% in our sample. *Minor* moves mostly meant a change of less than 2% in our sample. If a flat to *minor upward* movement is predicted, the investor pursues ASP1: writes covered call with strike close to the predicted price. If a *major upward* movement is predicted, the investor pursues ASP2: writes covered call with strike close to the predicted price. However, if the call premium is not large enough, the strategy is instead to buy shares and at-the-money call options and write calls with strike close to the predicted price. If a *minor downward* movement is predicted, the investor pursues ASP3: writes a covered call with strike close to the current stock price. If the price predictor indicates a *major downside* for a stock, the investor pursues ASP4: writes a covered call at a strike close to the predicted stock price and buys a put option with the strike near or above the closing stock price on the trade initiation day. The initial investment for the ASP strategies applied to a stock varies somewhat from \$10,000. This is because the number of shares was rounded to the nearest 100 as a CBOE equity option contract is for 100 shares.